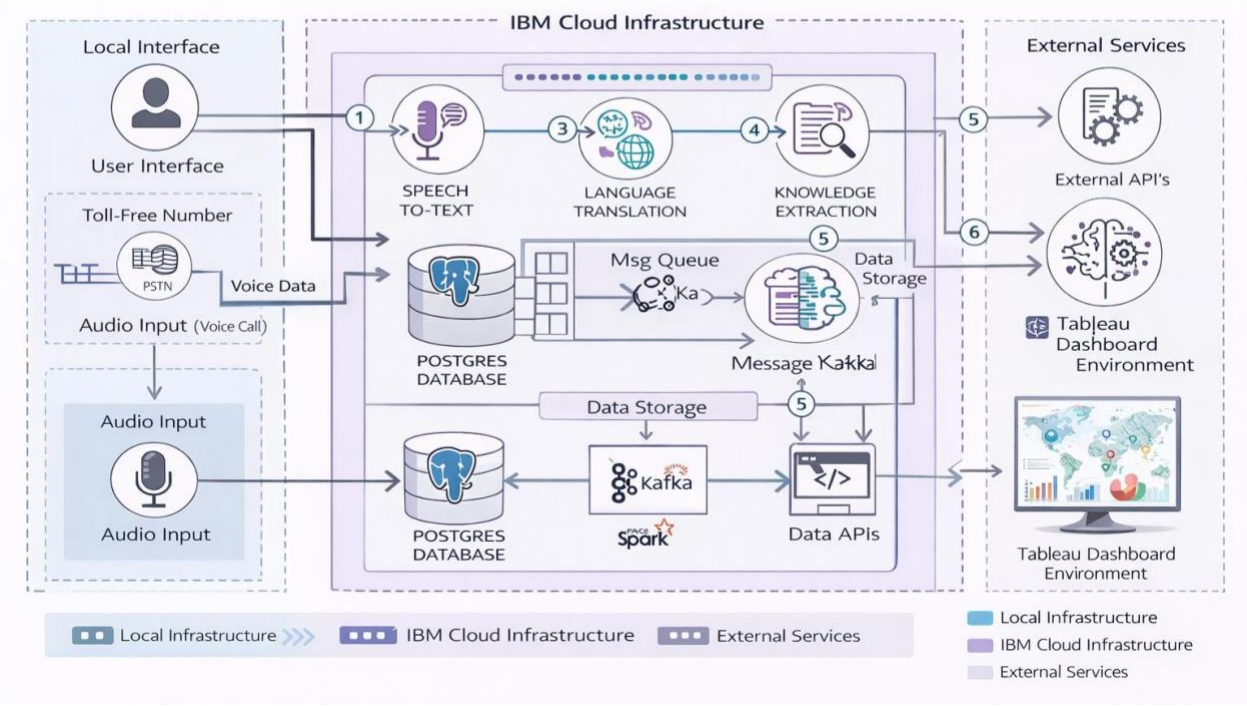
Project Design Phase – II

Technology Stack (Architecture & Stack)

Team Id	LTVIP2026TMIDS36716
Project Name	Heritage Treasures : An in-depth Analysis of UNESCO World Heritage sites using tableau

Technical Architecture :

The technical architecture of Heritage Treasures integrates UNESCO datasets and external APIs through an ETL process for data cleaning, transformation, and storage in a structured database. Tableau connects to the processed data to generate interactive dashboards, with optional machine learning models used for advanced insights and trend analysis.



S.No	Component	Description	Technology
1	User Interface	Dashboard interface for users to explore heritage site data and Insights	Tableau Dashboard, HTML, CSS
2	Application Logic-1	Data cleaning, preprocessing, and transformation logic	Python (Pandas, NumPy)
3	Application Logic-2	Data integration and ETL process	Python, Tableau Prep
4	Application Logic-3	Data analysis and aggregation logic	SQL Queries, Python
5	Database	Structured storage of UNESCO heritage dataset	MySQL / PostgreSQL
6	Cloud Database	Cloud-based data storage for scalability and remote access	IBM DB2 / IBM Cloudar
7	File storage	Storage of raw datasets (CSV, Excel, JSON files)	Local File System / IBM Cloud Object Storage
8	External API-1	UNESCO World Heritage Site data retrieval	UNESCO Open Data API
9	External API-2	Geo-location and map visualization support	Google Maps API / OpenStreetMap API
10	Machine Learning Model	Predictive analysis (trend prediction, risk classification of endangered sites)	Scikit-learn (Regression / Classification Models)
11	Infrastructure (Server/ Cloud)	Deployment of application and dashboards	Local Server / IBM Cloud / Cloud Foundry / Kubernetes

Application Characteristics :

S. No	Characteristics	Description	Technology
1	Open-Source Framework	Frameworks used for data processing and analysis	Python, Pandas, NumPy, Scikit-learn
2	Security Implementation	Data encryption, authentication, and secure access controls	SHA-256, SSL/TLS, IAM Controls
3	Scalable Architecture	Designed using modular ETL pipeline and cloud-based storage for scaling	3-Tier Architecture, Microservices, IBM Cloud
4	Availability	Ensures high availability using cloud hosting and backup systems	Load Balancer, Distributed Cloud Servers
5	Performance	Optimized queries, caching, and efficient data visualization	SQL Indexing, Tableau Extracts, CDN